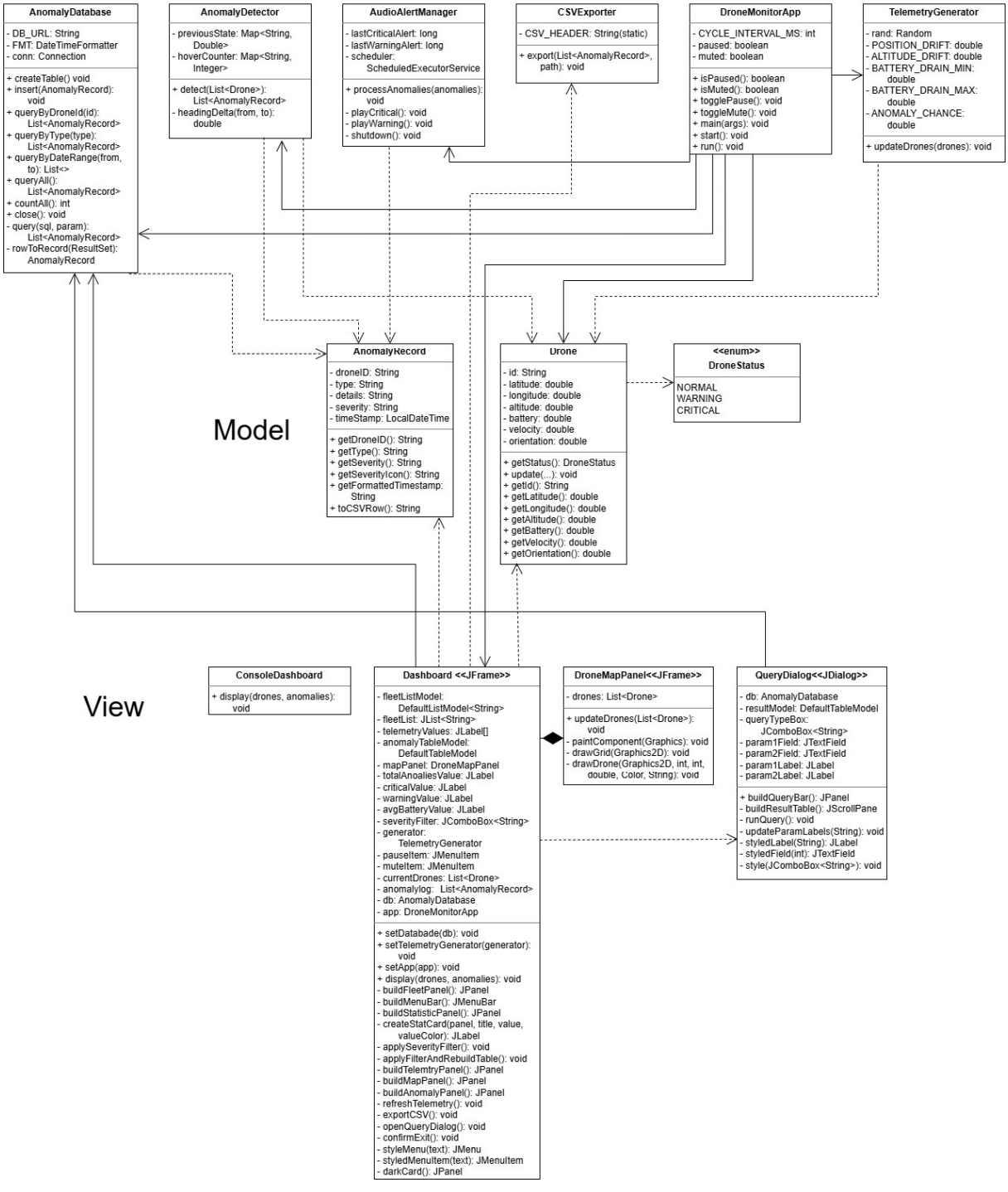
Controller


AnomalyDatabase
- DB_URL: String
- FMT: DateTimeFormatter
- conn: Connection
+ createTable() void
+ insert(AnomalyRecord): void
+ queryByDroneId(id): List<AnomalyRecord>
+ queryByType(type): List<AnomalyRecord>
+ queryByDateRange(from, to): List<>
+ queryAll(): List<AnomalyRecord>
+ countAll(): int
+ close(): void
- query(sql, param): List<AnomalyRecord>
- rowToRecord(ResultSet): AnomalyRecord
AnomalyDetector
- previousState: Map<String, Double>
- hoverCounter: Map<String, Integer>
+ detect(List<Drone>): List<AnomalyRecord>
- headingDelta(from, to): double
AudioAlertManager
- lastCriticalAlert: long
- lastWarningAlert: long
- scheduler: ScheduledExecutorService
+ processAnomalies(anomalies): void
- playCritical(): void
- playWarning(): void
- shutdown(): void
CSVExporter
- CSV_HEADER: String(static)
+ export(List<AnomalyRecord>, path): void
DroneMonitorApp
- CYCLE_INTERVAL_MS: int
- paused: boolean
- muted: boolean
+ isPaused(): boolean
+ isMuted(): boolean
+ togglePause(): void
+ toggleMute(): void
+ main(args): void
+ start(): void
+ run(): void
TelemetryGenerator
- rand: Random
- POSITION_DRIFT: double
- ALTITUDE_DRIFT: double
- BATTERY_DRAIN_MIN: double
- BATTERY_DRAIN_MAX: double
- ANOMALY_CHANCE: double
+ updateDrones(drones): void
Model
AnomalyRecord
- droneID: String
- type: String
- details: String
- severity: String
- timeStamp: LocalDateTime
+ getDroneID(): String
+ getType(): String
+ getSeverity(): String
+ getSeverityIcon(): String
+ getFormattedTimestamp: String
+ toCSVRow(): String
Drone
- id: String
- latitude: double
- longitude: double
- altitude: double
- battery: double
- velocity: double
- orientation: double
+ getStatus(): DroneStatus
+ update(...): void
+ getId(): String
+ getLatitude(): double
+ getLongitude(): double
+ getAltitude(): double
+ getBattery(): double
+ getVelocity(): double
+ getOrientation(): double
<<enum>>
DroneStatus
NORMAL
WARNING
CRITICAL
View
ConsoleDashboard
+ display(drones, anomalies): void
Dashboard <<JFrame>>
- fleetListModel: DefaultListModel<String>
- fleetList: JList<String>
- telemetryValues: JLabel[]
- anomalyTableModel: DefaultTableModel
- mapPanel: DroneMapPanel
- totalAnoaliesValue: JLabel
- criticalValue: JLabel
- warningValue: JLabel
- avgBatteryValue: JLabel
- severityFilter: JComboBox<String>
- generator: TelemetryGenerator
- pauseItem: JMenuItem
- muteItem: JMenuItem
- currentDrones: List<Drone>
- anomalylog: List<AnomalyRecord>
- db: AnomalyDatabase
- app: DroneMonitorApp
+ setDatabade(db): void
+ setTelemetryGenerator(generator): void
+ setApp(app): void
+ display(drones, anomalies): void
- buildFleetPanel(): JPanel
- buildMenuBar(): JMenuBar
- buildStatisticPanel(): JPanel
- createStatCard(panel, title, value, valueColor): JLabel
- applySeverityFilter(): void
- applyFilterAndRebuildTable(): void
- buildTelemtryPanel(): JPanel
- buildMapPanel(): JPanel
- buildAnomalyPanel(): JPanel
- refreshTelemetry(): void
- exportCSV(): void
- openQueryDialog(): void
- confirmExit(): void
- styleMenu(text): JMenu
- styledMenuItem(text): JMenuItem
- darkCard(): JPanel
DroneMapPanel<<JFrame>>
- drones: List<Drone>
+ updateDrones(List<Drone>): void
- paintComponent(Graphics): void
- drawGrid(Graphics2D): void
- drawDrone(Graphics2D, int, int, double, Color, String): void
QueryDialog<<JDialog>>
- db: AnomalyDatabase
- resultModel: DefaultTableModel
- queryTypeBox: JComboBox<String>
- param1Field: JTextField
- param2Field: JTextField
- param1Label: JLabel
- param2Label: JLabel
+ buildQueryBar(): JPanel
- buildResultTable(): JScrollPane
- runQuery(): void
- updateParamLabels(String): void
- styledLabel(String): JLabel
- styledField(int): JTextField
- style(JComboBox<String>): void